



HACKHAZARDS '26 HACKATHON SUBMISSION

HealthCurve Nova AI

Your intelligent, graph-powered multilingual healthcare companion.

GRAPH DATABASE

Neo4j AuraDB Connection

MULTILINGUAL OCR

Sarvam AI Integration

Meet Team Code Crafters

The innovative minds behind HealthCurve Nova AI, pushing boundaries in digital healthcare.



Aman Raj

Backend and development

Designed backend architecture, orchestrated API routing, and constructed Neo4j graph database pipelines.



Aditya Raj

Idea and frontend design

Engineered multilingual OCR integration using Sarvam AI and structured real-time LLM reasoning via Gemini.

Our Tech Stack



Sarvam AI

Multilingual OCR



Gemini 2.5 Flash

LLM
Engine



Neo4j AuraDB

Graph
Database



Python

Backend
Logic



Flask

REST API
Framework



HTML5

Structure



Vanilla CSS

Visual
Design



JavaScript

Interactivity

The Healthcare Challenge

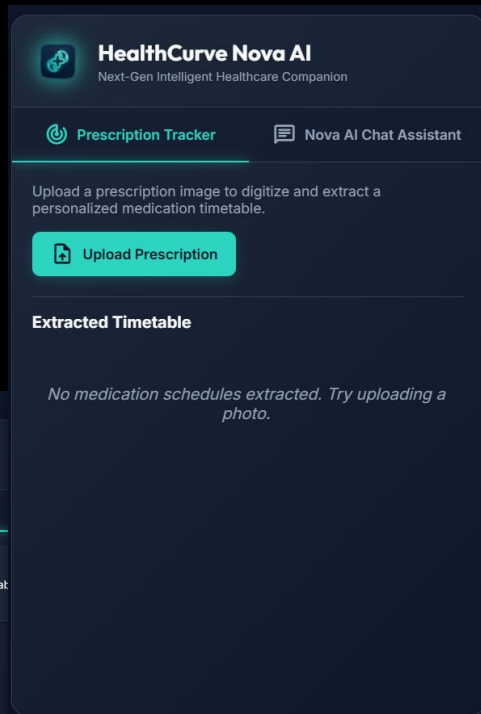
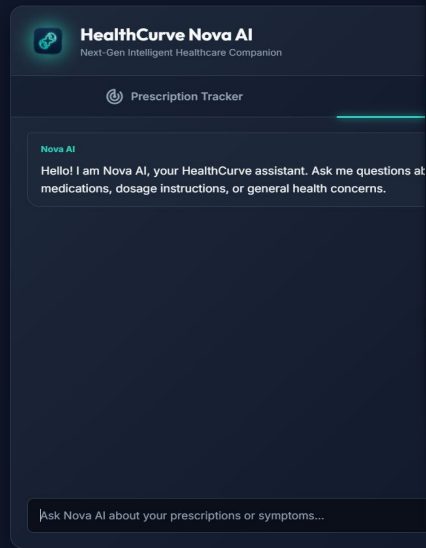
- **Unreadable Prescriptions:** Messy handwriting causes dosage errors.
- **Complex Schedules:** Tracking multi-drug treatments is exhausting.
- **Lack of Clarity:** No instant answers to medication questions.



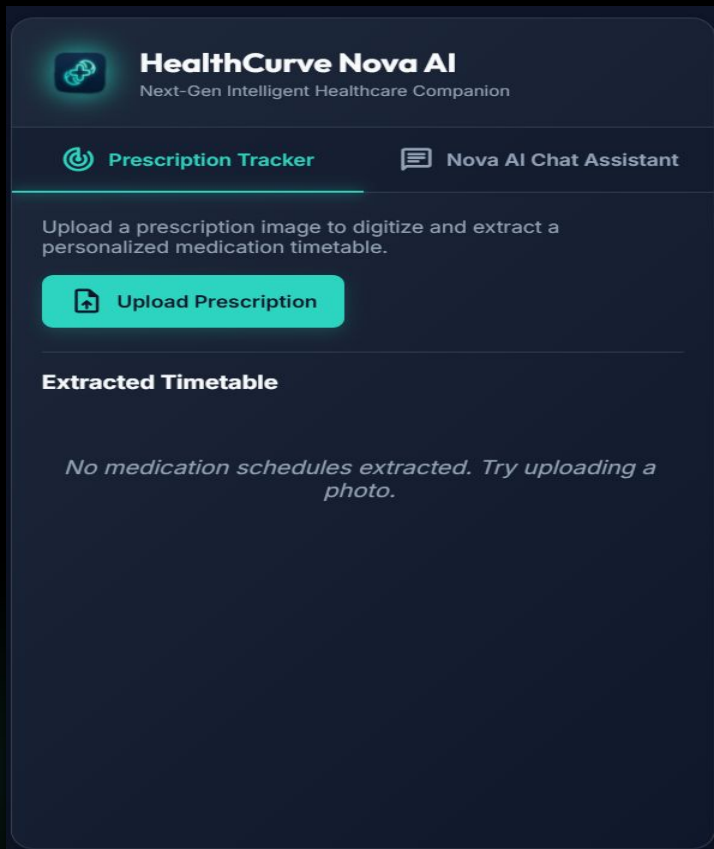
The Nova AI Solution

A next-gen assistant that digitizes, structures, and interacts with medical data in real-time.

- **Scan:** Instant digitization via Sarvam AI.
- **Structure:** Graph-mapping with Google Gemini.
- **Chat:** Conversational context-aware support.



Core Architecture



- **Frontend:** Glassmorphism SPA (HTML/CSS/JS).
- **Backend:** Python Flask API orchestrated with Gunicorn.
- **Intelligence:** Sarvam AI & Google Gemini 2.5 Flash.
- **Data:** Neo4j Graph Database (AuraDB).

Multilingual OCR

Breaking Language Barriers

Native support for **English + 22 regional Indian languages**.

Layout-aware text extraction ensures even messy handwritten prescriptions are parsed with clinical precision.



Graph-Powered Schedules

Raw text is converted to structured JSON and stored as a network of nodes.



Patient Node

The central identity anchor.



Medicine Node

Specific drugs and dosages.



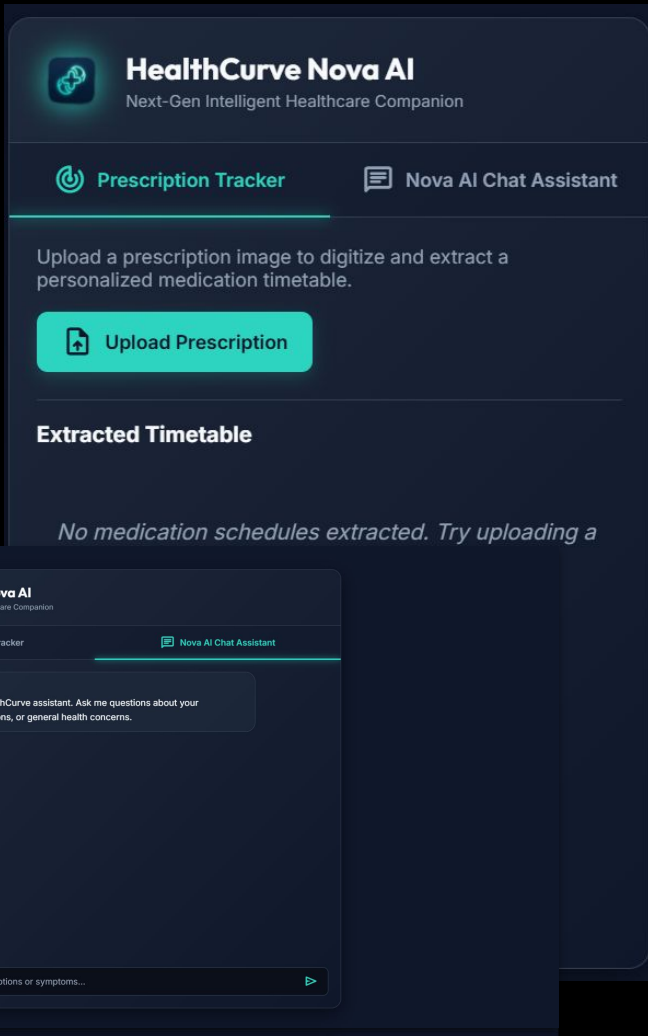
TimeSlot Node

Optimized hour-by-hour tracking.

Contextual Chat

Nova AI isn't just a bot—it's your medical records personified.

- **Memory:** Fetches active schedule from Neo4j.
- **Safety:** Provides dosage instructions contextually.
- **Voice:** Supports multiple regional dialects.



Technical Hurdles

- **Async Pipelines:** Implemented polling routines for Sarvam's job-based Document Intelligence endpoints.
- **API Stability:** Upgraded backend to target **Gemini-2.5-Flash** natively for 100% uptime.
- **Contextual Awareness:** Built Neo4j helper functions to feed live patient data into the LLM system prompt.

What's Next?

Adherence Tracking

"Taken" logs in Neo4j.

Push Reminders

Automated SMS/WhatsApp
alerts.

Interaction Checks

Drug-to-drug safety alerts.